Logic-Gated Biomaterials for Targeted Therapy: Exploring how *"Engineered modular biomaterial logic gates for environmentally triggered therapeutic delivery"* is Altering Delivery Methods for Bioactive Therapeutics

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I. Abstract

This paper provides an overview of the historical development of hydrogels, and presents an innovative development of logic-gated hydrogel release profiles. The history traces the evolution through three generations of hydrogels, highlighting their respective functionalities and applications. Emphasis is placed on smart hydrogels in the current era, showcasing their responsiveness to environmental stimuli for applications in advanced drug delivery and tissue engineering. Efforts to improve the efficacy of hydrogels with regard to targeted drug delivery has resulted in the introduction of a novel hydrogel system utilizing PEG and SPAAC cross-linking to design a logic-gated hydrogel cross-linking network. The technology incorporates a modular framework with 17 engineered cross-linkers, responding to various stimuli, and demonstrating precision in drug release. Applications in an in vitro cancer model showcase the system's efficiency, and a multifunctional hydrogel reveals its adaptability in controlled, sequential therapeutic agent release. Acknowledging limitations in proteolytic cleavage kinetics, the paper concludes by underscoring the groundbreaking potential of this hydrogel system. It heralds a new era in biomaterial science, offering versatile solutions in drug delivery, diagnostics, and regenerative medicine.

II. Historical Development of Hydrogels as Drug Delivery Systems

A. Definition

A hydrogel is a three-dimensional (3D) network of hydrophilic polymers that can hold a large amount of water while maintaining its structure due to the chemical or physical cross-linking of individual polymer chains (Bahram, M. et al). Hydrogels can be divided into 2 distinct categories based on the source of their constituent polymers: either natural or synthetic. Natural hydrogels are composed of collagen, fibrin, hyaluronic acid, matrigel, and derivatives of natural materials such as chitosan, and alginate fibers while synthetic hydrogels include poly (ethylene glycol) diacrylate, poly(acrylamide) and poly(vinyl alcohol). Each of these polymers has their own advantages as well as disadvantages. Natural polymers are easy to access, abundant, inexpensive, non-toxic, and biodegradable. Synthetic polymers have highly controllable physical and chemical properties in comparison to natural polymers and can be produced with long-chain structures as well as high molecular weight. However, synthetic polymers exhibit lower biological activity when compared to natural hydrogels.

B. Properties

In general, polymer-based hydrogels possess several beneficial characteristics such as the ability to mimic body tissues due to their high water content and soft consistency. Consequently, hydrogels are highly biocompatible. The swelling capacity of polymer-based hydrogels also enables them to absorb and retain significant amounts of biological fluids which creates a unique environment that facilitates essential biological interactions. The use of polymers within hydrogels plays an integral role in its functionality since hydrogels' ability to absorb water can be attributed to the hydrophilic functional groups attached to the polymeric backbone, while resistance to disintegration is due to cross-links between network chains.

C. Synthesis

Hydrogels are synthesized via a series of chemical reactions which include polymerization and parallel cross-linking of multifunctional monomers. Other methods include polymer-cross-linking agent reactions, multistep processes such as

polymer synthesis with reactive groups, followed by appropriate cross-linking with appropriate agents, and multistep processes such as polymer synthesis with reactive groups, followed by appropriate cross-linking with suitable agents.

D. Generations

The word “hydrogel” can be historically dated to an article published by Lee, Kwon and Parkin in 1894. However, the material described was not a hydrogel as we know it today; it was instead a colloidal gel made with inorganic salts. Hydrogels, as we now know them, were the first materials developed for permanent contact applications inside the human body and have actually existed for more than half a century. In 1949, the first-ever hydrogel reported for biomedical implantation was Ivalon, poly(vinyl alcohol) cross-linked with formaldehyde. In 1960, another hydrogel, Polyhydroxyethylmethacrylate (PHEMA) was developed by Wichterle and Lim for applications in soft contact lenses, which brought significant attention to the hydrogel market. The main features of their design included shape stability and softness similar to that of the soft surrounding tissue, chemical and biochemical stability, absence of extractables, and high permeability for water-soluble nutrients and metabolites. Subsequently, in the following years, research into the biomedical applications of hydrogels began to increase. Broadly, the history of hydrogels can be divided into 3 eras. The first generation (1960s) of hydrogels was designed to create a material with high swelling and good mechanical properties based on simple chemically crosslinked networks of synthetic polymers. These hydrogels were comprised of polymers of alkene monomers subjected to radical-induced chain addition reactions, covalently cross-linked hydrophilic polymers, and cellulose-based hydrogels. The second generation of hydrogels (starting in the 1970s) was inspired by the works of Katchalsky and characterized by the ability to respond to particular stimuli including changes in temperature, pH, and biomolecules to achieve a specified function (e.g. gel formation or drug release). This generation consisted mainly of PEG/ polyester block copolymers. Temperature-sensitive hydrogels display a phase transition from gel to sol state from low to high temperatures while pH-sensitive hydrogels are hydrolyzed at high or low pH environments respectively. Biomolecule-sensitive hydrogels exhibited conformational changes in the presence of specific biomolecules. Finally, the third generation (mid-1990s) of hydrogels focused heavily on "crosslinking". This generation was particularly characterized by the investigation and development of stereo-complexed materials. This offered the ability to enhance and fine-tune the mechanical, thermal, and degradation properties of polymers. This generation comprised hydrogels such as PEG-PLA (Rizwan, M. et al, 2017).

E. Present-Day Smart Hydrogels

Over the past two decades, the field of hydrogel technology has undergone significant evolution, leading to the emergence of "smart hydrogels" in the 2010s. These advanced hydrogels have laid a robust foundation for innovative applications in areas such as drug delivery systems and tissue engineering. Smart hydrogels are distinguished from their predecessors by their remarkable ability to respond to environmental stimuli, including changes in pH, temperature, electrical and magnetic fields, light, and the concentration of biomolecules. This responsiveness enables them to undergo dramatic and reversible transitions, both in volume and from gel to sol states, in response to minimal alterations in specific environmental conditions.

In the realm of drug delivery, smart hydrogels have been a focal point of extensive research, offering groundbreaking strategies to revolutionize pharmacotherapy. The unique properties of these hydrogels allow for a more controlled and targeted drug release, enhancing the efficacy of drug delivery systems. By responding to physiological variables, smart hydrogels can modulate drug pharmacokinetics, ensuring that drugs are released at the desired rate and location, thus optimizing therapeutic outcomes. This ability to respond to specific environmental triggers enables a more precise control over drug absorption, reducing potential toxicity and improving the overall safety profile of therapeutic interventions.

Furthermore, smart hydrogels have shown immense potential in tissue engineering, where their biocompatibility and tunable physical properties make them ideal candidates for creating scaffolds that mimic the natural extracellular matrix. This adaptability facilitates the development of tailored tissue constructs, promoting cell growth and differentiation in a

controlled manner. The application of smart hydrogels in tissue engineering opens up new avenues for regenerative medicine, offering promising solutions for tissue repair and regeneration.

In summary, the advent of smart hydrogels marks a significant advancement in biomaterial science, offering versatile and dynamic platforms for a range of biomedical applications. Their ability to respond to environmental stimuli not only enhances the performance of drug delivery systems but also paves the way for innovative approaches in tissue engineering, making them a cornerstone of next-generation medical therapies.

F. Biomaterial of Interest

The hydrogel system investigated for the development of logic-gated devices relies on the strategic use of poly(ethylene glycol) (PEG) hydrogels, employing specific cross-linking chemistry known as strain-promoted alkyne–azide cycloaddition (SPAAC) (Badeau et al., 2018). In the context of this study, the unique properties of SPAAC-based PEG hydrogels, as detailed in a supplemental article, play a pivotal role in achieving reproducibility, controlled swelling behavior, and overall suitability for the intended device.

To address the challenge of batch-to-batch irreproducibility inherent in hydrogel formulations, a study employed dendritic polymer end-groups. Another paper introduces the use of SPAAC for cross-linking PEG hydrogels, focusing on dendron-functionalized PEG chains. The SPAAC process facilitates rapid gelation without the need for additional reagents or catalysts, offering a controlled and efficient method for forming hydrogels. This aligns with the requirements of logic-gated devices, where precise and reproducible hydrogel properties are essential for reliable performance (Badeau et al., 2018).

The polymer of choice, PEG, is selected for its well-established biocompatibility, hydrophilicity, and ease of functionalization. The incorporation of dendritic end-groups, specifically azide-terminated first- and second-generation dendrons enhances the structural control of the hydrogel (Badeau et al., 2018). The dendritic macromolecules not only contribute to the stability of the hydrogel network but also ensure a well-defined number of end groups at the periphery, promoting consistency between batches (Hodgson et al., 2017).

Considering the specific properties of SPAAC-based PEG hydrogels, supplemental articles provide insights into hydrogel swelling kinetics. The study reveals distinct differences in the swelling behavior of first- and second-generation dendron-functionalized hydrogels. The G1 hydrogels exhibit increased swelling over time, reaching their highest swelling ratios after 53.5 hours. On the other hand, G2 hydrogels display minimal swelling, particularly at higher concentrations(Hodgson et al., 2017). This controlled swelling behavior is advantageous for biomedical applications, ensuring that the physical and mechanical properties of the hydrogel remain stable over time, a critical aspect for logic-gated devices.

Reproducibility is a key consideration for the development of logic-gated hydrogel devices. The second paper establishes the reproducibility of SPAAC-based PEG hydrogels by comparing Young’s modulus values between different batches of precursor polymers. The dendritic hydrogels, particularly G1 and G2, exhibit consistent mechanical properties, confirming the reliability of the hydrogel fabrication method. This reproducibility is essential for creating reliable logic-gated systems with consistent performance (Hodgson et al., 2017).

III. Current Clinical Challenges

A. Limitations

Several clinical problems confront the existing landscape of hydrogel-based therapeutic delivery systems, necessitating cautious consideration for future developments. Despite significant advances in medicinal development and cell engineering, several limitations in the clinical application of hydrogels remain.

One of the key issues is the requirement for accurate and targeted therapeutic administration to disease locations. While hydrogels are versatile, they may have limitations in terms of particular localization within the body. The prevalence and often unpredictability of diseases present a substantial challenge in ensuring that medicinal payloads reach their intended sites. This constraint is especially important in cases when diseases have several symptoms or where the target areas are difficult to reach. It is critical to improve the precision and accuracy of hydrogel-based delivery systems to maximize therapeutic outcomes while minimizing off-target effects (Badeau et al., 2018).

Furthermore, present technologies for hydrogel generation and medication loading entail processes that may be hazardous or pose potential dangers. For example, using UV photopolymerization or chemical cross-linking procedures necessitates the careful removal of hazardous chemicals before hydrogel implantation (Hoare & Kohane, 2008). This complicates the therapeutic application of hydrogels since guaranteeing total clearance of hazardous residues can be challenging. Strategies for addressing the toxicity concerns associated with hydrogel production are critical for expanding their therapeutic relevance. Furthermore, the preformed nature of some hydrogels, which are prepared outside the body before implantation, may limit their adaptability to different anatomical contexts. The need for approaches that facilitate in-situ gelation, reduce the risk of premature gelation, and improve hydrogel adaptability to varied clinical settings, has become evident (Hoare & Kohane, 2008).

Additionally, the present duration of drug release from hydrogels may not be adequate for long-term therapeutic interventions. High water-content hydrogels frequently demonstrate relatively fast drug release characteristics over short periods. This constraint contrasts with the ability of other delivery technologies, such as microspheres or macroscopic devices made of hydrophobic polymers, to provide longer release profiles. Improving the duration of drug release from hydrogels is critical, especially for applications requiring long-term therapeutic benefits (Hoare & Kohane, 2008). Modulating the rate of drug delivery over time or creating easily tunable on-off release dynamics can greatly expand the scope of hydrogel-based therapeutics.

B. Improvement Potential

Addressing these issues could lead to many clinical breakthroughs and revolutionize the landscape of hydrogel-based medicinal delivery. greater precision in targeting specific disease locations would not only enhance treatment efficiency but would also reduce the danger of off-target consequences, adding to greater patient safety (Hoffman, 2013). The development of solutions to reduce the toxicity associated with hydrogel production procedures would allow for more seamless clinical translation. The capacity to achieve in-situ gelation, which is responsive to the specific properties of diverse anatomical regions, has the potential to revolutionize the adaptability and usability of hydrogel-based systems (Badeau et al., 2018).

Furthermore, advances in establishing extended drug release from hydrogels would open up new avenues for their application in chronic diseases and long-term therapeutic interventions. Tailoring hydrogel characteristics to allow for the transport of sensitive molecules such as proteins, antibodies, or nucleic acids while preserving their biological activity is an important route for future research (Hoffman, 2013). Overcoming these current constraints is the key to realizing the full promise of hydrogel-based therapeutic delivery systems, providing creative solutions to complicated clinical challenges and considerably improving patient outcomes.

IV. Next-Step Proposal

A. Current Problem of Targeted Therapeutic Delivery

The technology presented in the article aims to solve a critical problem in the field of drug and cell-based therapeutics: the effective and targeted delivery of treatments to specific diseased sites. This challenge is pivotal in the development of clinical therapies, particularly considering the widespread and often unknown locations of disease sites. Traditional methods, while employing polymer-based vehicles to facilitate delivery and protect therapeutic cargo, often face limitations such as immune recognition, clearance, non-specific cellular uptake, and the need to mimic native tissue environments for cell-based therapies.

The specific issue addressed by the new technology is the lack of precision in targeting disease sites due to the non-uniqueness of individual biomarkers. For instance, characteristics like matrix metalloproteinase (MMP) activity, reducing conditions, and subphysiological pH, which are targeted in cancer microenvironments, are also present in healthy tissues like joints and the stomach. This leads to suboptimal selectivity in therapeutic delivery. Existing materials that respond to single or dual environmental factors do not provide enough specificity, and redesigning these materials for each new application is not synthetically feasible due to constraints on material composition and vehicle geometry. Additionally, current degradable materials for live cell release are limited to single biological inputs, restricting the potential of advanced cellular therapeutics.

B. Overview of Technology

To overcome these challenges, the article introduces a novel modular chemical framework for hydrogels, which uses multiple environmental cues to trigger user-programmable Boolean logic for precise control over material degradation and therapeutic delivery. This approach promises higher efficiency, specificity, and versatility in therapeutic delivery, applicable to drug delivery, diagnostics, and regenerative medicine. The hydrogels developed can sense and respond to a combination of disease-associated biochemical markers, offering a solution to the problem of targeted delivery and expanding the possibilities for precision medicine. The creation of highly precise and adaptable hydrogel-based delivery systems capable of releasing therapeutics exclusively at disease sites by responding to multiple, specific pathophysiological cues, thereby allowing for the overcoming of the limitations surrounding existing delivery methods in terms of targeting accuracy and specificity.

The article describes an innovative technology involving the development of multi-stimuli-responsive hydrogel platforms, designed for precise and controlled therapeutic delivery. These hydrogels are equipped with biocomputation capabilities, enabling them to sense various biological inputs and execute complex Boolean logic-based algorithms for controlled therapeutic release. This functionality not only enhances the specificity of drug release but also overcomes limitations in traditional drug and cell delivery methods. Furthermore, another key feature of these hydrogels is their modular design, which allows for the customization of inputs and logic functions, thereby creating a diverse array of materials each with unique release properties. The hydrogels are constructed using cytocompatible bioorthogonal chemistries, ensuring they can be safely formed and degraded in the presence of living cells. Integral to these hydrogels are stimuli-sensitive components within synthetic cross-linkers, which react with polymer macromers to form hydrogels with precise molecular architectures. The implementation of logic gates is a standout aspect, with simple YES gates realized through single stimuli-labile moieties, and more complex OR and AND gates achieved by arranging multiple degradable units in series or parallel configurations. The covalent cleavage of these linkers leads to material degradation and the release of any encapsulated or tethered therapeutic agents. Furthermore, the concept of expandable logic allows for the engineering of hydrogels capable of responding to additional dynamic stimuli by integrating multiple gates into a single logic circuit. This advancement in hydrogel technology signifies a major leap forward in therapeutic delivery, offering a new paradigm for creating materials that are structurally simple yet functionally complex, and holds immense potential in fields like drug delivery, diagnostics, and regenerative medicine.

C. Results and Implications

With reference to the study, the team successfully engineered 17 unique cross-linkers, each sensitive to different combinations of enzymes, reductants, and light. This achievement was made possible through the innovative use of three basic logic gates: YES, OR, and AND shown in Figure 2. The cross-linkers were designed with regions sensitive to specific stimuli, known as stimuli-labile moieties (red and blue regions in the figure). In the YES-gate cross-linkers, a single stimuli-labile region was included, leading to material degradation upon interaction with its corresponding chemical input. The OR-gate and AND-gate cross-linkers, however, incorporated two such regions. For the OR-gate, these regions were arranged in series, allowing material degradation upon the presence of either stimulus. In contrast, the AND-gate's design required both stimuli, as its regions were connected in parallel, for complete material degradation. To confirm the molecular responsiveness of these cross-linkers, the researchers utilized Matrix-Assisted Laser Desorption/Ionization Time-of-Flight (MALDI-TOF) mass spectrometry. This analysis validated the precise reaction of the materials to the specified environmental stimuli at a molecular level.

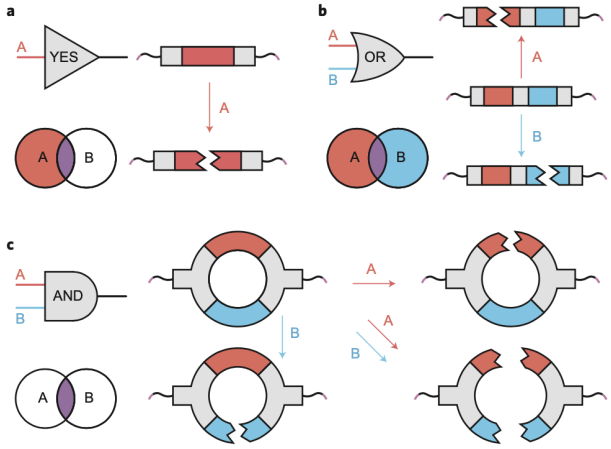


Figure 2: Diagram of the YES, OR, and AND crosslinkers (Badeau et al., 2018)

A significant application of these hydrogels was demonstrated in an in vitro cancer model. The R∧E hydrogels were engineered to release the chemotherapeutic agent doxorubicin (DOX) in response to specific stimuli associated with cancer microenvironments, thus inducing population-wide apoptotic death in plated cervical cancer-derived HeLa cells as shown in Figure A1 in the appendix. This targeted delivery approach showed that, in the absence of treatment or the presence of a single stimuli, normal cell proliferation was observed with a dsDNA content of 95 ± 3% and 97 ± 2%, respectively. When both relevant stimuli were present, there was a marked reduction in cancer cell viability, as indicated by the drastically reduced dsDNA content of 1.8 ± 0.2%, demonstrating the effective release of the therapeutic agent.

Moreover, the study confirmed the viability of cells encapsulated within these hydrogels post-release, regardless of the stimuli applied as shown in Figure 3. This finding underscores the potential of these engineered hydrogels as advanced platforms for live-cell delivery, offering promising implications for biomedical applications.

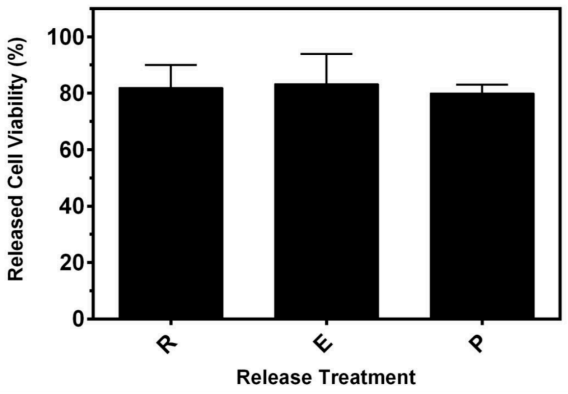


Figure 3: Graph showing encapsulated cell viability post release (Badeau et al., 2018)

The researchers also developed a multifunctional hydrogel, incorporating three distinct logic regions, each marked by a unique fluorophore. These specialized regions are fine-tuned to undergo degradation only upon encountering their respective triggers, enabling a highly selective response mechanism. This precision in control allows for the hydrogel to exhibit specific reactions or material breakdowns in targeted areas, facilitating complex, multi-step processes as shown in Figure A2 in the appendix. Consequently, this design enables the sequential release of various types of cells or drugs, each from its designated region within the hydrogel. This approach is especially beneficial in biomedical applications, where controlled and targeted delivery of multiple therapeutic agents or cells is of paramount importance. Such a system offers a significant advancement in the field, providing a versatile platform for precise therapeutic delivery tailored to specific needs and conditions.

Lastly, the study explored the distinct mechanical properties of the hydrogels. Experiments involving the treatment of the E∧P linker with various stimuli combinations before incorporation into polyethylene glycol (PEG) revealed insightful findings. Untreated linkers resulted in a stiff hydrogel material, as evidenced by a high storage modulus (G′) of 1660 ± 170 Pa. Additionally, when treated with either an enzyme or light, the material retained its robust gel structure (G′ = 1580 ± 130 Pa and 1540 ± 110 Pa, respectively). However, simultaneous treatment with both stimuli prevented gel formation, highlighting a major alteration in the material's structure with a much lower G′ of 200 ± 30 Pa. The results show that the shear modulus (a measure of material stiffness) scales with the density of crosslinks within the material. The data suggest that the mechanical properties of the hydrogels depend solely on the final logic state of the Boolean linker. In this case, the logic state is determined by whether the linker is intact or cleaved (which depends on the exposure to enzyme, light, or both).

D. Remaining Limitations

The new technology described in the article, while demonstrating significant potential, does face certain limitations. One notable constraint pertains to the proteolytic cleavage kinetics of the matrix metalloproteinase (MMP)-degradable substrates, which can be reduced in the presence of internal ring strain. This issue was evident in the case of higher-order, three-input cross-linkers used as complex logic operators for controlling material degradation. In some instances, these cross-linkers did not fully degrade, a phenomenon attributed to the strained substrates' slower proteolytic cleavage. Despite this limitation, the overall approach proved robust, with 132 of the 136 treatment conditions successfully resulting in engineered degradation (defined as either complete degradation or less than 30% nonspecific release). This achievement underscores the potential of the technology in achieving complex biomaterial computation with high fidelity through the combination of simple YES/OR/AND logic gates. Looking forward, the modular framework's initial success suggests the possibility of substituting the chosen stimuli-labile groups with other chemically orthogonal moieties sensitive to various factors like pH, additional proteases, visible light, temperature, or ultrasound, potentially overcoming current limitations.

V. Conclusion

This report emphasizes the transformative potential of hydrogel technology in drug delivery and tissue engineering. Our investigation focuses on "smart hydrogels," specifically a novel modular chemical framework enabling intricate biocomputation. These hydrogels exhibit a remarkable capacity to sense and respond to biological inputs, executing precise Boolean logic-based algorithms. Key findings highlight their ability to undergo targeted degradation and controlled therapeutic release, showcasing specificity and adaptability to diverse stimuli. Our research explores the synthesis and efficacy of hydrogels, revealing their intricate design with YES, OR, and AND logic gates tailored for specific stimuli. Promising results in cancer therapeutics, especially in the targeted release of doxorubicin, showcase the precision of these hydrogels. The study affirms their suitability for live-cell delivery and biomedical applications, despite limitations such as decreased proteolytic cleavage kinetics and incomplete degradation under specific conditions. Overall, the hydrogels demonstrate robustness, with a high success rate in achieving engineered degradation across most conditions.

As we look ahead, the prospect of substituting stimuli-labile groups with chemically orthogonal moieties sensitive to a wider range of environmental factors opens new avenues for further refinement. Enhancing the precision, efficiency, and versatility of these hydrogels could revolutionize their application in drug delivery and tissue engineering. The potential for improvements, such as extending drug release duration, adapting to varied clinical settings, and ensuring the safe clinical translation of hydrogels, underscores the significance of ongoing research in this field. In conclusion, the advancements in smart hydrogel technology represent a significant leap forward in biomaterial science, offering dynamic and versatile platforms for biomedical applications. As we continue to address the existing challenges and explore potential enhancements, the full promise of hydrogel-based therapeutic delivery systems is on the horizon, poised to provide innovative solutions to complex clinical challenges and significantly improve patient outcomes.

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VI. Appendix

Figure A1: Diagram representing the functioning of R∧E hydrogels to release DOX (Badeau et al., 2018)

Figure A2: Diagram representing the sequential and spatiotemporally varied delivery (Badeau et al., 2018)